

Supervised Learning Study for Income Prediction and Wine Quality Classification

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Abstract—In this study, four different supervised learning algorithms on two benchmark classification tasks were implemented. These tasks included income prediction using the Adult Income dataset and multiclass quality prediction using the Wine Quality dataset. Before implementation, hypothesis were driven on which algorithms will perform best; comparing between decision trees, k-nearest neighbors, support vector machines, and neural networks. These algorithms were implemented under a reproducible protocol that analyzes learning behavior, model complexity, and runtime efficiency. The results provided insight on the model trade-offs under class imbalance, feature sparsity, and nonlinear structures.

I. INTRODUCTION

Supervised learning methods are widely used for classification and regression tasks across many data types including structured data with mixed features, sparsity, noise, and data imbalance. Although many algorithms can achieve competitive predictive performance, their behavior varies significantly depending on dataset properties such as dimensionality, non-linearity, imbalance etc. Understanding these differences is important for model selection.

This report presents a hypothesis-driven empirical study on two benchmark classification datasets with contrasting characteristics. The first, *Adult Income*, is a binary classification task with mixed categorical and numerical features and significant class imbalance. The second, *Wine Quality*, is a multiclass classification problem with continuous features and imbalance across quality labels. Together, these datasets provide complementary settings for analyzing how model families respond to feature structure and imbalance.

Four supervised learning algorithms are trained, tuned, and evaluated under a consistent and reproducible dataset. Beyond final predictive performance, learning curves, model-complexity trends, confusion matrices, and runtime efficiency are also analyzed to diagnose bias–variance behavior and computational tradeoffs. Neural networks are implemented using both scikit-learn and PyTorch under stochastic gradient descent to enable comparisons of optimization dynamics.

The primary objective is to explain **why** certain models perform better in specific settings and **how** dataset characteristics influence model behavior. By basing empirical results in explicit hypotheses derived from exploratory data analysis and learning theory, this study provides practical insights into supervised model selection under realistic constraints.

II. DATASETS AND EXPLORATORY DATA ANALYSIS

The two datasets that are explored for this assignment include the Adult dataset and the Wine dataset. An Exploratory Data Analysis (EDA) was carried out in order to derive hypotheses before implementing predictive models for it.

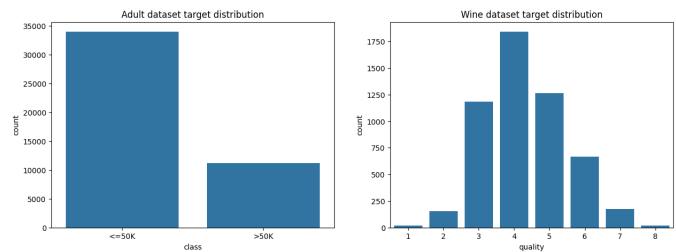


Fig. 1: Target class distributions for the Adult Income and Wine Quality datasets. The Adult dataset shows significant class imbalance, with substantially more counts of the $\leq 50K$ class. The Wine dataset exhibits a multiclass distribution centered around mid-range quality ratings, with fewer counts in extreme classes - which indicates imbalance across quality levels.

A. Adult Income Dataset

Looking at the Adult dataset, it consists of 45,222 data entries with 15 columns. After dropping the duplicate rows (47 duplicate rows were found) in this dataset, it contains 45,175 rows. The dataset consists of 6 numerical and 9 categorical columns. All columns are non-null which indicate no missing values in the dataset for any of the features. The following list gives a deeper analysis on the numeric variables:

- **Age:** The age ranges from 17 to 90 years with a mean of about 38.5 years and a standard deviation of 13.2 years. The median age is 37 years.
- **Fnlgwt:** This consists of a wide range and mean of about $1.89e+05$, which suggests a non-normal distribution. Skewness of 1.45 indicates that the data is skewed towards the right.
- **Education-Num:** This represents the years of education and ranges from 1 to 16, with a mean of 10.12. Skewness of -0.31 indicates that the data is slightly skewed towards the left.

- **Capital-Gain and Capital-Loss:** These features have a large number of zeros, indicating that majority of the individuals have no capital gains or losses while a few do have significant values. Skewness of 11.45 and 4.51 respectively, indicates that the data is skewed heavily towards the right.
- **Hours-Per-Week:** This column ranges from 1 to 99 hours, with a mean of 40.94 hours and concentrated around 40 hours. Skewness of 0.34 indicates that the data is slightly skewed towards the right.

Meanwhile, categorical columns provide the following information:

- **Workclass:** The most frequent variable in 'Workclass' is 'Private' with 7 unique values.
- **Education:** 'HS-grad' is the most common variable with 16 unique values.
- **Marital-status:** 'Married-civ-spouse' is the most frequent marital status with 7 unique values.
- **Occupation:** This column consists of 14 unique values.
- **Race:** 'White' is the most frequent race with 5 unique values.
- **Sex:** 'Male' is the most frequent sex with 2 unique values.
- **Country:** 'United States' is the most common native country with 41 unique values.

Figure 1 depicts the target variable for this dataset, 'class', with two categorical variables. 33,973 instances belong to '<=50K' (75.2%) and 11,202 instances belong to '>50K' (24.8%), indicating an imbalanced dataset.

Overall, the Adult dataset looks relatively clean with no missing values and few duplicates. However, it does exhibit significant skewness in several numerical features, specially in capital gain, capital loss, and final weight features. Furthermore, the target variable is class 'imbalanced' which is important to consider during modeling.

B. Wine Quality Dataset

The wine dataset consists of 6947 rows and 14 columns with 1177 duplicate rows (indicating 5320 unique rows). Furthermore, of the 14 columns, 5320 rows have non-null values, indicating no missing entries. The numerical columns data is summarized as follows:

- **Fixed_acidity:** This column ranges from 3.8 to 15.9, with a mean of 7.22. Most values for this column are concentrated between 6.4 and 7.7. The data is also skewed towards the right by 1.65.
- **Volatile_acidity:** Ranges from 0.08 to 1.58, with a mean of 0.34. The data is spread out pretty considerably but shows a right skew of 1.5.
- **Citric_acid:** Ranges from 0 to 1.66, with a mean of 0.32. Analyzing the data shows that many of the wines have low or no citric acid. The data is slightly skewed towards the right by 0.48.
- **Residual_sugar:** Ranges from 0.6 to 65.8, with a high mean of 5.44, but a small median of 3, suggesting a strongish right skew of 1.71.

- **Chlorides:** Ranges from 0.009 to 0.611 with a mean of 0.056. Furthermore, the values for this column are generally low. However, skewness of this column shows that the data is highly skewed towards the right with a value of 5.34.
- **Free_sulfur_dioxide:** Ranges from 1 to 289, with a mean of 30.53, which shows a wide range with skewness towards the right (1.36).
- **Total_sulfur_dioxide:** Ranges from 6 to 440, with a mean of 115.74, which also indicates a broad distribution. The data appears to be near symmetric with a very minor right skew of 0.06.
- **Density:** The density ranges from 0.987 to 1.039, with a mean of 0.9947 and slight skew towards the right of 0.67.
- **pH:** Ranges from 2.72 to 4.01, with a mean of 3.22 and centered around an acidic range for wine. It is slightly skewed to the right by 0.39.
- **Sulphates:** Ranges from 0.22 to 2, with a mean of 0.53. The data is skewed right by 1.81.
- **Alcohol:** Ranges from 8 to 14.9, with a mean of 10.49. The data is slightly skewed to the right by 0.55.
- **Class:** Discrete numerical feature with a mean of 4.31. The data is slightly right skewed by 0.31.
- **Type:** Binary numerical feature (0 or 1), with a mean of 0.75. It is skewed slightly left by -1.12.
- **Quality:** The target variable, with values ranging from 1 to 8, and a mean of 4.31. The most common qualities were 3, 4, and 5. Overall, quality is slightly right skewed by 0.31.

The target variable, quality, shown in figure 1, is an ordinal variable with values ranging from 1 to 8. The most frequent value from the quality ratings is 4, with 1841 occurrences (34.61%), followed by 5 (23.8%), and 3 (22.27%). Meanwhile, the lowest ratings belong to 6, 7, 2, 1, and 8. The mean quality is about 4.31, with a standard deviation of 1.17, which shows that the majority of quality scores fall within the mid range.

III. HYPOTHESES

A. Hypothesis for Adult Dataset

After analyzing the Adult dataset, the following hypotheses were formed for each algorithm:

- **Decision Trees:** Expected to perform well due to their ability to handle mixed numerical and categorical features with minimal preprocessing [1], [2]. Since splits are feature-based rather than distribution-based, trees are relatively robust to skewness [1]. Class imbalance can be addressed through weighting or resampling techniques such as SMOTE [3].
- **K-Nearest Neighbors (KNN):** Expected to perform moderately well but require extensive preprocessing. As a distance-based lazy learner [4], KNN is sensitive to scaling and skewed features, meaning transformations and normalization will be critical. Class imbalance may bias predictions toward the majority class unless resampling or weighting is applied.

- **Support Vector Machines (SVM):** Expected to perform moderately well with proper preprocessing. SVMs rely on numerical inputs and are sensitive to feature scaling, particularly with nonlinear kernels [5]. Skewed features must be scaled appropriately, and imbalance should be handled via class weights or resampling.
- **Neural Networks (NNs):** Expected to perform well with sufficient preprocessing and architecture selection [6]. Numerical encoding and scaling are necessary for convergence. Performance may degrade under class imbalance unless weighted losses or resampling strategies are used.

B. Hypothesis for Wine Dataset

For the Wine dataset, the hypotheses are as follows:

- **Decision Trees:** Expected to perform strongly due to purely numerical features and the ability to model nonlinear splits. Trees naturally support multiclass classification and are relatively robust to skewness. Imbalance can be mitigated using weighting or sampling.
- **K-Nearest Neighbors (KNN):** Expected to perform moderately well with proper scaling. As a distance-based method, it remains sensitive to skewness and class overlap. Multiclass voting is supported, but performance may decline if class boundaries are not clearly separated.
- **Support Vector Machines (SVM):** Expected to behave similarly to KNN but potentially improve with nonlinear kernels. Proper scaling is essential to prevent skewed features from negatively impacting optimization.
- **Neural Networks (NNs):** Expected to perform well due to their ability to learn complex nonlinear relationships. With appropriate scaling and regularization, strong multiclass performance is anticipated.

IV. METHODOLOGY

A. Data Splitting and Evaluation Protocol

Both the Adult and Wine datasets were stratified and split into train–test sets (80/20). Stratification preserved the original class distributions. The training set was used for tuning and cross-validation, while the test set was reserved strictly for final evaluation.

Five-fold cross-validation was used for all models except KNN (three-fold due to computational cost). Model performance was evaluated using:

- **F1-score:** For the Adult dataset (binary classification).
- **Macro-F1:** For the Wine dataset to account for multiclass imbalance [7].

Accuracy was reported but not used for model selection due to imbalance. Learning curves, model-complexity curves, and runtime measurements were also generated to analyze bias–variance behavior and computational efficiency under a consistent evaluation framework.

B. Pre-processing and Leakage Controls

To prevent data leakage, all preprocessing steps were implemented within scikit-learn pipelines. Categorical features

(Adult dataset) were one-hot encoded, while numerical features were standardized. Transformations were fit only on training folds within cross-validation.

Duplicate rows were removed prior to splitting to avoid potential information leakage. Hyperparameter tuning and preprocessing were performed exclusively on training data. A dummy classifier baseline was included to verify that all learned models outperformed trivial predictors.

C. Model Configuration and Tuning

Four supervised algorithms were evaluated: Decision Trees, KNN, SVM, and Neural Networks.

- **Decision Trees:** Tuned using `max_depth` and `ccp_alpha`.
- **KNN:** Tuned over the number of neighbors using brute-force distance computation.
- **SVM:** Evaluated with linear and RBF kernels; regularization parameter C was swept logarithmically. Class weights were used for Adult imbalance.
- **Neural Networks:** Implemented with stochastic gradient descent; hidden-layer size and L2 regularization were tuned to analyze model capacity and generalization.

V. RESULTS

A. Adult Income Results

For the Adult Income dataset, Decision Trees (Figure 2) perform best at a moderate depth value, however, increasing tree depth increases the transition into overfitting since the training and F1 scores widen. Furthermore, Cost Complexity Pruning (CCP) also confirms this behaviour because increasing the value off `ccp` reduced variance but, with excessive pruning resulted in underfitting (shown by declining performance). Lastly, the learning curve shows convergence between training and validation scores as the sample size increases, as both converge at approximately 0.65 F1 score. However, additional training data lowers performance improvements, showing a moderate bias–variance tradeoff.

TABLE I: Performance of k-nearest neighbors on the Adult income dataset

Metric	Value
Best k	11
Accuracy	0.835
F1-score (> 50K)	0.641
Cross-validation F1 (mean)	0.647
Training F1 (mean)	0.698
Fit time (s)	0.136
Prediction time (s)	18.60

The k-nearest neighbors models (Figure 3) show that it performed worst than Decision Trees on the Adult dataset. Looking at model complexity, the model had significant overfitting at small k values (high training but low validation scores). But, as the k value was increased, the predictions were more in line with training, but did cause more bias since there were more k values. Furthermore, although Table 1 shows good accuracy, it insignificant due to data imbalance, and the

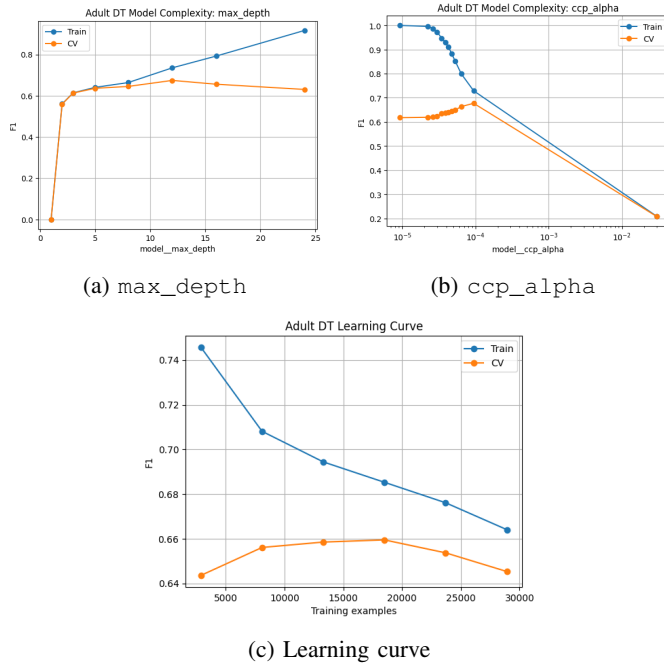


Fig. 2: Curves representing Decision Tree performance on the Adult Income dataset. (a) Model complexity as a function of tree depth (`max_depth`). Performance improves at low depths but overfitting happens beyond a depth 12 as training F1 scores increase while cross-validation (CV) F1 scores decline. (b) Cost-complexity pruning via `ccp_alpha`. Increasing pruning strength reduces variance, but too much pruning ($> 10^{-4}$) does lead to underfitting as indicated by the declining training scores. (c) Learning curve showing convergence of training and validation F1 as sample size increases, with both curves stabilize near a F1 score ≈ 0.65 , which indicates limited improvements from additional data and moderate bias–variance tradeoff.

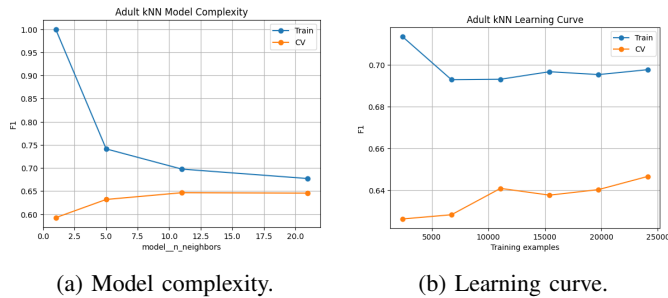


Fig. 3: k-Nearest Neighbors (kNN) performance on the Adult dataset. (a) Model complexity analysis shows significant overfitting at low values of k , where training F1 approaches 1.0 while cross-validation (CV) performance remains substantially lower. Increasing k reduces variance and improves generalization, with the highest validation performance observed near $k = 11$. (b) The learning curve indicates that performance improves as training size increases, but, a consistent gap between training and validation curves indicates moderate variance and limited gains from additional data.

prediction time was substantially higher compared to decision trees (3-5 seconds), which strengthen the initial hypothesis that decision trees outperform KNN (due to distance calculations in high dimensional space). Lastly, learning curves confirmed that kNN struggled to generalize very effectively even after increasing training data.

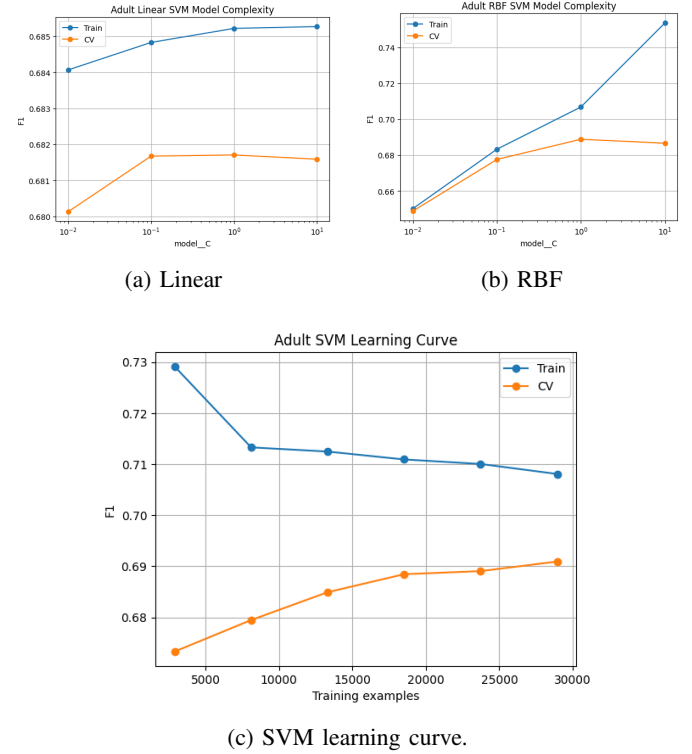


Fig. 4: Support Vector Machine (SVM) analysis on the Adult dataset. (a) The Linear kernel shows minimal performance improvement as C increases. (b) For the RBF kernel, performance improves as C increases up to 10^0 (1.0), where the cross-validation (CV) F1-score reaches its peak. Beyond this point, the training score continues to increase while the CV score begins to decline, indicating the onset of overfitting as regularization weakens. (c) The learning curve shows convergence between training and validation F1-scores as sample size increases, suggesting moderate variance and limited gains from additional training data. Overall, $C = 1.0$ provides the best balance between bias and variance.

Analyzing the results from Support Vector Machines (SVMs), Figure 4, they demonstrate similar performance compared to KNN models. Linear SVMs achieved somewhat stable results but since they do not capture nonlinear feature interactions effectively, their results were suboptimal. Furthermore, although RBF SVMs achieved higher validation F1-scores, they also increased the computational cost close to KNN, due to distance calculations. Lastly, Learning curves indicated relatively stable generalization performance but with slightly diminishing returns as training size increased.

The last algorithm for "Adult" dataset that was implemented was Neural Networks (Figure 5). A Hidden unit sweep was

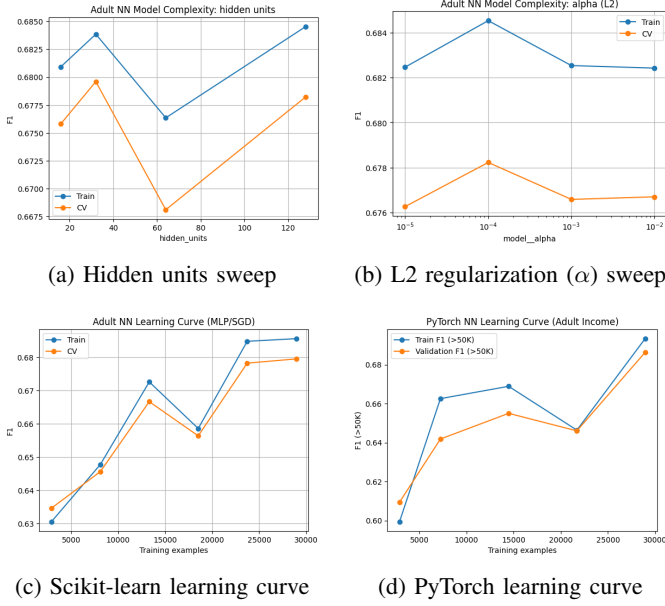


Fig. 5: Neural Network performance analysis. (a) Model complexity with varying hidden units shows peak validation F1 near 32 units, with insignificant returns beyond that size. (b) L2 regularization sweep indicates optimal performance at $\alpha = 10^{-4}$, with stronger regularization leading to slight underfitting. (c) scikit-learn MLP learning curve demonstrates convergence of training and validation F1 around 0.68, suggesting moderate bias and steadying variance. (d) PyTorch MLP learning curve show steady improvement with increasing data, indicating stable generalization behavior.

carried out to find the optimal parameters, and it revealed that moderate model capacity yielded optimal validation performance, with 32 hidden units achieving the highest cross-validation F1-score. Increasing capacity beyond this point led to minimal improvement in training performance and also slightly widened the train-validation gap, indicating mild overfitting. Additionally, regularization sweeps showed limited sensitivity to α , with $\alpha = 10^{-4}$ producing the strongest validation performance. Lastly, learning curves indicate a moderate bias-variance tradeoff. As training size increased, validation F1 steadily improved while the gap between training and validation curves narrowed. However, gains began to plateau at larger sample sizes, indicating diminishing returns from additional data. Implementing NN using Pytorch also showed improved learning with more training examples. Nevertheless, training time was higher than decision trees and kNN likely due to iterative gradient optimization. However, prediction time remained low which does make NNs computationally efficient.

The confusion matrix demonstrates that the results are mostly inline with the initial hypotheses. Decision tree and NN results suggest that they outperform KNN and SVM with respect to precision and computational efficiency. Furthermore, implementing a dummy classifier helps prove that the imple-

TABLE II: Adult Dataset Confusion Matrix Summary Across Models

Model	Correct	Total	Errors	Summary
Decision Tree	7686	9045	1359	Low-moderate errors
kNN ($k = 11$)	7559	9045	1486	Moderate-high errors
SVM (RBF)	7289	9045	1756	More false positives
NN (MLP/SGD)	7705	9045	1340	Best overall balance
Dummy	6803	9045	2242	Only predicts ≤ 50

mented models are learning correctly and they are meaningful.

To summarize further, the results show meaningful learning but they are far from perfect. This is because, the "Adult" dataset contains high-dimensional one-hot encoded categorical features, which increase model complexity and optimization difficulty. Despite this, neural network do demonstrate competitive minority-class F1 performance compared to other models. The results suggest that models such as Decision Tree and Neural Networks based ones, with moderate capacity networks and effective regularization/ hidden layers, can generalize effectively even in high-dimensional sparse feature spaces.

B. Wine Quality Results

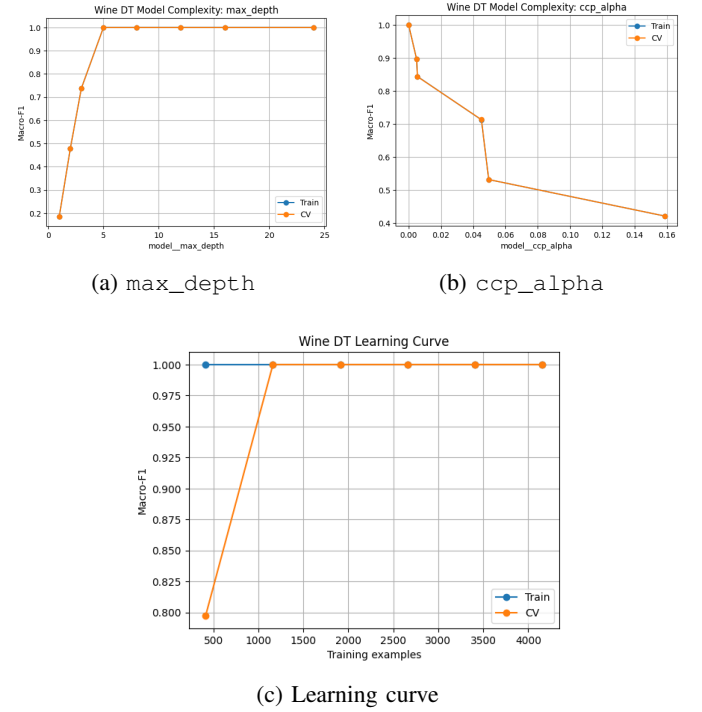


Fig. 6: Decision Tree analysis on the Wine dataset. (a) Model complexity as a function of max_depth, showing quick convergence to perfect Macro-F1 by depth 5. (b) Effect of cost-complexity pruning (ccp_alpha), showing that increasing pruning uniformly reduces performance. (c) Learning curve demonstrating rapid convergence of validation performance to 1.0, indicating low bias and low variance.

For the Wine dataset, the Decision Tree model results in (Figure 6) suggests that the underlying structure of the

dataset is easily separable using axis-aligned splits. The cost-complexity pruning results further confirm this behavior. As `ccp_alpha` increases, performance declines uniformly for both training and validation sets, which indicates that pruning primarily reduces model capacity rather than correcting overfitting. The minimal gap between the curves however, across all depths and pruning values, suggests low variance. Lastly, the learning curve reinforces the conclusion, as cross-validation performance quickly converges to 1.0 with relatively few training examples. The early stabilization of both curves indicates that the model is not data or variance limited. Overall, the Wine dataset appears much easier than the Adult dataset and is highly structured/ easily captured by a Decision Tree classifier.

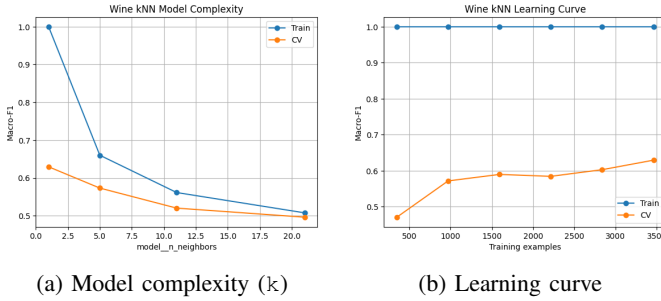


Fig. 7: k-Nearest Neighbors (kNN) analysis on the Wine dataset. (a) Model complexity curve showing the relationship between the number of neighbors (k) and Macro-F1 score. Small values of k lead to severe overfitting, but larger values smooth predictions but increase bias. (b) Learning curve illustrating persistently high variance with training performance remains near perfect while cross validation performance improves slightly but plateaus well below training, indicating limited generalization.

For the Wine dataset, kNN showed clear signs of high variance and overfitting. As seen in Figure 7(a), small values of k resulted in very high training performance but substantially lower cross-validation performance, indicating strong overfitting. Increasing k reduced variance by smoothing the decision boundary, however, this also increases bias.

The learning curve in Figure 7(b) further supports this behavior. Although validation performance improves as more training data is added, there is also a noticeable and consistent gap between training and cross validation scores. This suggests that even with additional data, the model struggles to learn in a meaningful fashion, which also highlights the limitations of instance-based learning for this dataset.

SVMs performed particularly well due to the low-dimensional, continuous feature space. Model-complexity curves showed that moderate regularization yielded optimal performance, while excessive model capacity led to overfitting. Learning curves indicated that validation performance improved steadily with additional data, suggesting data as the limiting factor.

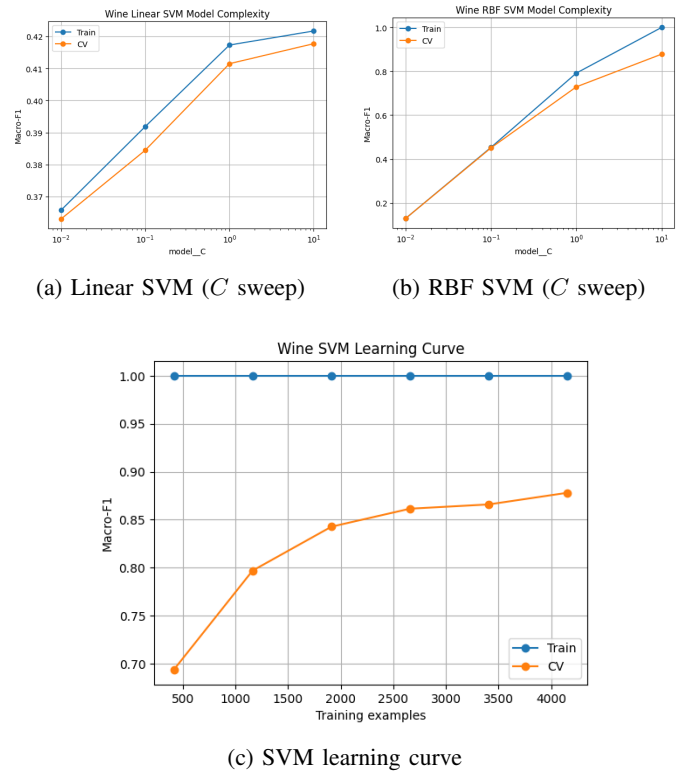


Fig. 8: Support Vector Machine analysis on the Wine dataset. (a) For the Linear SVM, Macro-F1 improves as the regularization parameter C increases from 10^{-2} to 10^0 and plateaus near $C = 10^1$, with a small and consistent training-CV gap indicating stable generalization. This also shows that the wine dataset does not have complex non linear patterns that the linear kernel is unable to learn. (b) For the RBF kernel SVM, increasing C substantially improves performance, with training performance reaching 1.0 while cross-validation reaches a maximum near 0.88. The gap widens at higher C values indicating overfitting as model flexibility increases. (c) The learning curve shows that while training performance remains perfect, cross-validation performance improves steadily with additional data, suggesting that the model continue to improve from larger training sets.

For the Wine dataset, modest architectures yielded improved validation performance. The best configuration used 36 hidden units. However, regularization strength had minimal impact across tested values, suggesting that overfitting was not severe in this dataset.

The best scikit-learn MLP achieved:

- Accuracy: 0.976
- Macro-F1: 0.689

Despite very high accuracy, Macro-F1 remained moderate due to strong class imbalance. Upon a deeper inspection of the results, the model performs well on majority quality classes but struggles on the minority classes. Furthermore, learning curves show steady improvement as training size increases. Using PyTorch training and validation curves converge gradually,

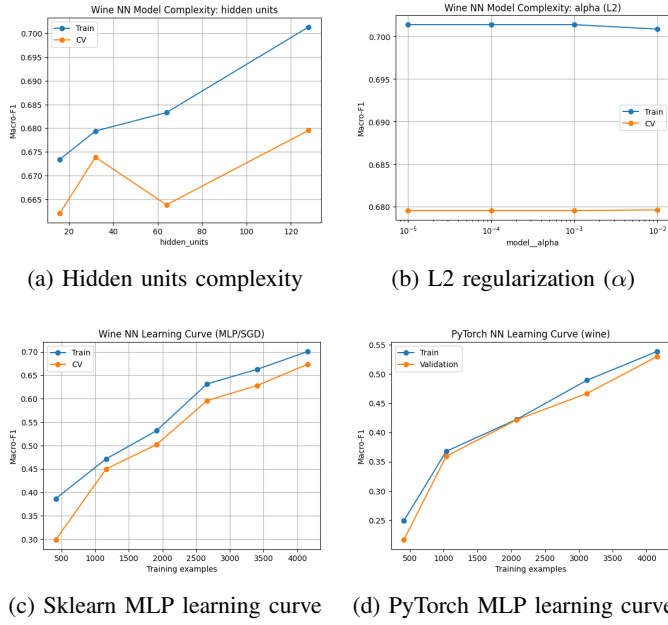


Fig. 9: Neural Network analysis on the Wine dataset. (a) Increasing `hidden_units` improves training performance gradually, while cross-validation (CV) performance peaks at 36 units and declines at higher complexity, indicating overfitting beyond moderate capacity. However, it does increase beyond about 64 layers at the expense of runtime. (b) The L2 regularization parameter (α) shows stable behavior across several magnitudes, with only slight underfitting at higher regularization strengths. The results show little to now impact on the Macro-F1 score. (c) The sklearn MLP learning curve demonstrates steady improvement in Macro-F1 as training size increases, with a consistent but small generalization gap. (d) The PyTorch implementation exhibits similar learning behavior, which confirm that performance trends are driven by the NN architecture rather than framework.

indicating improving generalization with additional data and architecture control.

TABLE III: Wine Dataset Confusion Matrix Summary Across Models

Model	Correct	Total	Errors	Comments
Decision Tree	1300	1300	0	Perfect classification
kNN	1163	1300	137	Moderate misclassification
SVM (RBF)	1291	1300	9	Near-perfect performance
NN (MLP/SGD)	1265	1300	35	Strong generalization
PyTorch NN	1247	1300	53	Slightly lower Macro-F1
Dummy	450	1300	850	Predicts majority class only

Table III summarizes the confusion matrix results across all models for the Wine dataset. The Decision Tree achieved perfect classification, correctly predicting all 1300 samples, indicating that the dataset patterns are easily separable under axis-aligned splits. This does seem a little suspect though, but it seems like the patterns are suited well for decision trees to work well. The RBF SVM also demonstrated near-perfect performance with only 9 misclassifications, confirming

strong nonlinear separation capability. The neural network models (both Scikit and PyTorch) performed competitively, with moderate errors but strong overall learning potential. In contrast, kNN showed noticeably higher misclassifications despite perfect training behavior earlier in complexity analysis, reinforcing the high variance tendency. Lastly, the Dummy classifier highlights the impact of class imbalance, as predicting only the majority class resulted in substantial error. To summarize, the hypotheses were in line with the results, since the "Wine" dataset was easier to model compared to the "Adult" dataset. The Decision Tree model performed best followed by SVM with RBF kernel, which was a little surprising as it required less complexity than expected.

VI. DISCUSSION

The experimental results largely supported the initial hypotheses. Decision Trees performed strongly across both datasets due to their ability to model nonlinear relationships and handle mixed feature types, but deeper trees required regularization to prevent overfitting. kNN did not work great, due to the curse of dimensionality, struggling on the high dimensional Adult dataset and incurring significant prediction time cost. SVMs provided strong generalization performance, particularly with the RBF kernel which was able to account for non-linearity. Lastly, Neural networks were competitive across both datasets but did require more tuning and training time.

Table IV summarizes final test performance using the required metrics: Accuracy and F1 for Adult ($> 50K$ as the positive class), and Accuracy and Macro-F1 for Wine.

TABLE IV: Final test performance summary. Adult reports F1 for the $> 50K$ class. Wine reports Macro-F1.

Model	Adult Acc	Adult F1 ($> 50K$)	Wine Acc	Wine Macro-F1
Decision Tree	0.850	—	1.000	1.000
kNN	0.836	0.642	0.895	0.624
SVM (RBF)	0.806	0.689	0.993	0.947
NN (sklearn MLP/SGD)	0.852	0.672	0.976	0.689
PyTorch NN	—	—	0.959	0.635
Dummy	0.752	0.000	0.346	0.064

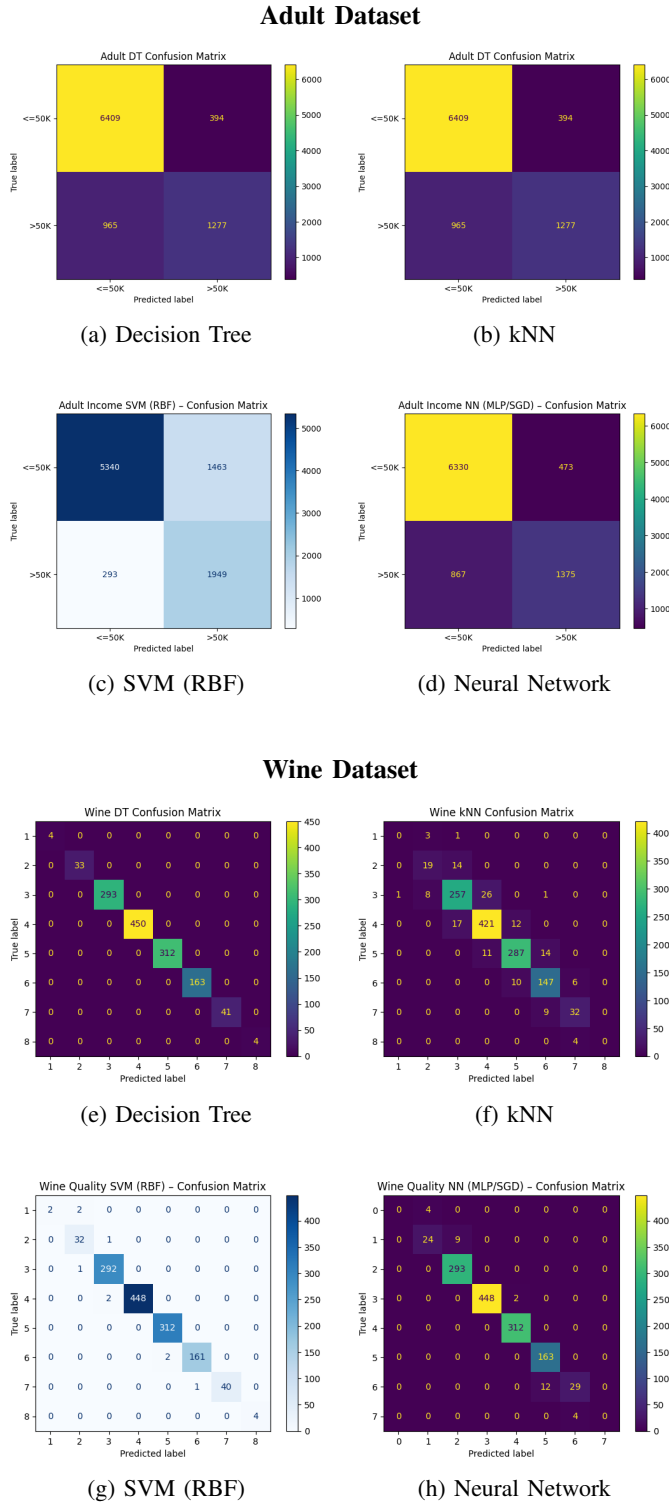
Runtime results aligned with expectations. Decision Trees were fast to train and predict. kNN showed a major prediction time constraint on "Adult" dataset due to instance-based inference in a high-dimensional one-hot feature space. Furthermore, neural networks required longer training time but maintained efficient inference once trained.

TABLE V: Wall-clock runtime (seconds). Executed locally in Jupyter.

Model	Adult fit	Adult predict	Wine fit	Wine predict
Decision Tree	0.461	0.049	0.024	0.005
kNN	0.145	18.479	0.007	0.042
NN (sklearn)	9.498	0.067	1.659	0.008

Confusion matrices reinforce that accuracy alone is insufficient for imbalanced classification. On Adult, the Dummy model achieved high accuracy by predicting only the majority class ($\leq 50K$) but completely failed on the minority class ($> 50K$). Among trained models, SVM (RBF) showed the

Fig. 10: Confusion matrices for the final selected configuration of each algorithm on both datasets.



strongest minority-class recall, while Decision Trees and kNN exhibited more false negatives.

For Wine, most errors occurred between adjacent quality labels (e.g., 3–4, 4–5, 6–7), consistent with ordinal overlap. Decision Trees achieved perfect classification on this split, while SVM (RBF) was near perfect with minor boundary errors. Lastly, kNN showed greater confusion across neighboring classes.

Per-class analysis revealed that extreme minority classes (1 and 8) were the most difficult to predict. The PyTorch neural network, in particular, failed to predict these rare classes, resulting in near-zero recall. In contrast, mid-range classes (between 3-6) were learned well across most models.

VII. CONCLUSION

This study compared four supervised learning algorithms across two classification tasks with different structural characteristics. The results show that model performance depends heavily on dataset properties, and that accuracy alone is not sufficient for evaluation, especially under class imbalance. Using F1-score, learning curves, confusion matrices, and runtime analysis provided clearer insight into bias–variance tradeoffs and computational behavior.

Decision Trees and SVMs showed the most consistent performance across both datasets, while NNs showed promising results. Pruning helped control variance for trees, while SVM (particularly RBF) achieved strong generalization as it captured non linear patterns. Neural Networks performed competitively, especially on the Adult dataset, though improvements on the Wine dataset were not drastic. Model complexity analysis showed diminishing returns beyond moderate capacity, suggesting that simply increasing complexity does not always improve performance.

Overall, model selection should stability, efficiency, and robustness to imbalance aside from accuracy. Future work may include ensemble approaches or more advanced neural architectures to further improve minority class prediction.

AI USE STATEMENT

AI tools (ChatGPT and Visual Studio Code Copilot) were used for brainstorming structure, clarifying requirements, and assisting with minor code debugging/refactoring.

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